

PART 1: Shortest Distance from a point to a function

Given $y = f(x) = x^2 + 5$ & five points $(0,0); (-4,0); (-8,0); (2,0); (6,0)$

1) Set Up the Distance Function

for a point, $P = (x_0, y_0)$

point on parabola: $Q = (x, f(x)) = (x, x^2 + 5)$

Euclidean Distance: $d(x) = \sqrt{(x-x_0)^2 + (f(x)-y_0)^2}$

* minimize squared distance $D(x) = d(x)^2$

$$\therefore D(x) = (x-x_0)^2 + (x^2+5-y_0)^2$$

* all points have $y_0 = 0$

$$\therefore D(x) = (x-x_0)^2 + (x^2+5)^2$$

POINT (-4,0):

$$2x^3 + 11x - (-4) = 0$$

$$2x^3 + 11x + 4 = 0$$

$$x^* = \sqrt[3]{-1 + \sqrt{\frac{16}{16} + \frac{1331}{216}}} + \sqrt[3]{-1 - \sqrt{\frac{16}{16} + \frac{1331}{216}}}$$

$$x^* \approx -0.355469693$$

$$y = x^2 + 5 = (-0.355469\ldots)^2 + 5$$

$$y \approx 5.126359$$

$$\therefore Q \approx (-0.355470, 5.126359)$$

$$\text{Distance: } d = \sqrt{(-0.355469693+4)^2 + (5.126359)^2}$$

$$d \approx 6.28945$$

POINT (-8,0):

$$2x^3 + 11x + 8 = 0$$

$$x^* \approx -0.672078127$$

$$y = x^2 + 5 = (-0.672\ldots)^2 + 5$$

$$y \approx 5.451689$$

$$\therefore Q \approx (-0.672078, 5.451689)$$

$$\text{Distance: } d = \sqrt{(-0.672078+8)^2 + (5.451689)^2}$$

$$d \approx 9.133420$$

2) Analytical Solution

$$D'(x) = 2(x-x_0) + 2(x^2+5)(2x)$$

$$= 2x - 2x_0 + 4x^3 + 20x$$

$$D'(x) = 4x^3 + 22x - 2x_0$$

$$4x^3 + 22x - 2x_0 = 0$$

$$2x^3 + 11x - x_0 = 0$$

POINT (2,0):

$$2x^3 + 11x - 2 = 0$$

$$x^* \approx 0.180744604$$

$$y = x^2 + 5 = (0.18074\ldots)^2 + 5$$

$$y \approx 5.032668$$

$$\text{Distance: } d = \sqrt{(0.180744604-2)^2 + (5.032668)^2}$$

$$d \approx 5.351396$$

POINT (6,0):

$$2x^3 + 11x - 6 = 0$$

$$x^* \approx 0.519903661$$

$$y = x^2 + 5 = (0.519903661)^2 + 5$$

$$y \approx 5.270300$$

$$\therefore Q \approx (0.519903661, 5.270300)$$

$$\text{Distance } d = \sqrt{(0.519903661-6)^2 + (5.270300)^2}$$

$$d \approx 7.603125$$

3) Verifying Critical Point is a Minimum

$$D''(x) = 12x^2 + 22$$

$$12x^2 + 22 > 0$$

$$\therefore D''(x) > 0 \quad \forall x$$

* Using Cardano's Formula for exact analytical soln

$$x = \sqrt[3]{-\frac{x_0}{2} + \sqrt{\left(\frac{x_0}{2}\right)^3 + \left(\frac{22}{3}\right)^3}} + \sqrt[3]{-\frac{x_0}{2} - \sqrt{\left(\frac{x_0}{2}\right)^3 + \left(\frac{22}{3}\right)^3}}$$

b/c there is only one real soln $D''(x) = 12x^2 + 22 > 0 \quad \forall x$

$\therefore D(x)$ is strictly convex & the critical point is the global minimum

$$\therefore x^* = \sqrt[3]{\frac{x_0}{4} + \sqrt{\frac{x_0^2}{16} + \frac{1331}{216}}} + \sqrt[3]{\frac{x_0}{4} - \sqrt{\frac{x_0^2}{16} + \frac{1331}{216}}}$$

↳ analytical solution for every point

$$y^* = (x^*)^2 + 5$$

$$d_{\min} = \sqrt{(x^*-x_0)^2 + (y^*)^2}$$

POINT (0,0):

$$2x^3 + 11x = 0$$

$$x^* = 0$$

corresponding point on parabola: $(0,5)$

$$\text{Distance: } d = \sqrt{(0-0)^2 + (5-0)^2}$$

$$d = 5$$

also $D''(0) = 22 > 0 \quad \therefore$ this is the minimum

Summary of Analytical Results

POINT	Closest x	Closest point on Parabola	Minimum Distance
(0,0)	0.00000	(0, 5.00000)	5.00000
(-4,0)	-0.355470	(-0.355470, 5.126359)	6.28945
(-8,0)	-0.672078	(-0.672078, 5.451689)	9.133420
(2,0)	0.180745	(0.180745, 5.032668)	5.351396
(6,0)	0.519904	(0.519904, 5.270300)	7.603125

PART 1A : Newton-Raphson by Hand

$$D'(x) = 2(x - x_0) + 2(f(x) - y_0)f'(x)$$

$$D''(x) = 2 + 2[f'(x)]^2 + 2(f(x) - y_0)f''(x)$$

$$f(x) = x^2 + 5$$

$$f'(x) = 2x$$

$$f''(x) = 2$$

$$\therefore D'(x) = 2(x - x_0) + 2(x^2 + 5)(2x)$$

$$D'(x) = 4x^3 + 22x - 2x_0$$

$$D''(x) = 2 + 2(2x)^2 + 2(x^2 + 5)(2)$$

$$= 2 + 8x^2 + 4x^2 + 20$$

$$D''(x) = 12x^2 + 22$$

$$\text{Newton-Raphson: } x_{n+1} = x_n - \frac{D'(x_n)}{D''(x_n)}$$

$$\therefore x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

$$\Rightarrow x_{n+1} = x_n - \frac{2x_n^3 + 11x_n - x_0}{6x_n^2 + 11}$$

* Will use x_0 (guess) = 0 as the initial guess for every point
& tolerance 10^{-7}

Point (0,0)

$$x_1 = 0; \text{ \& since } D'(0) = 0$$

$$x^* = 0$$

(-4, 0)

Iteration	x_n
0	0
1	-0.363636
2	-0.38548194
3	-0.355461693
4	-0.355469693

$$(-8, 0) : x_1 = -\frac{8}{11} = -0.727273$$

Iteration	x_n
0	0
1	-0.727273
2	-0.672992
3	-0.672078
4	-0.672078

$$(2, 0) : x_1 = \frac{2}{11} = 0.181818$$

Iteration	x_n
0	0
1	0.181818
2	0.180745
3	0.180744604
4	0.180744604

$$(6, 0) : x_1 = \frac{6}{11} = 0.545455$$

Iteration	x_n
0	0
1	0.545455
2	0.520068
3	0.519904
4	0.51990366

Point (2, 0) ← Full worked example

$$D'(x) = 4x^3 + 22x - 4$$

$$D''(x) = 12x^2 + 22$$

Iteration 1: $x_0 = 2$

$$D'(2) = 4(2)^3 + 22(2) - 4 = 32 + 44 - 4 = 72$$

$$D''(2) = 12(4) + 22 = 48 + 22 = 70$$

$$x_1 = 2 - \frac{72}{70} = 2 - 1.02857 = 0.97143$$

Iteration 2: $x_1 = 0.97143$

$$D'(0.97143) = 4(0.97143)^3 + 22(0.97143) - 4 = 21.03832166$$

$$D''(0.97143) = 12(0.97143)^2 + 22 = 33.65716$$

$$x_2 = 0.97143 - \frac{21.038}{33.657} = 0.97143 - 0.6314 = 0.34003$$

Iteration 3: $x_2 = 0.34003$

$$D'(0.34003) = 4(0.34003)^3 + 22(0.34003) - 4 = 0.1573 + 7.4807 - 4 = 3.638$$

$$D''(0.34003) = 12(0.34003)^2 + 22 = 1.387 + 22 = 23.387$$

$$x_3 = 0.34003 - \frac{3.638}{23.387} = 0.34003 - 0.1556 = 0.18443$$

very close to true answer (0.180745)
after just 3 iterations

Newton Point $(-4, 0)$

it.1 $D'(0) = 4(0)^3 + 22(0) - 2(-4) = 8$

$$D''(0) = 22$$

$$\therefore x^{(1)} = 0 - \frac{8}{22}$$

$$x^{(1)} = -0.3636364$$

it.2 $D'(x^{(1)}) = 4(-0.3636364)^3 + 22(-0.3636364) + 8$
 ≈ -0.19233689

$$D''(x^{(1)}) = 12(-0.19233689)^2 + 22$$

$$\approx 23.5897686$$

$$\therefore x^{(2)} = -0.3636364 - \frac{-0.19233689}{23.5897686}$$

$$x^{(2)} = -0.355481939$$

it.3 $D'(-0.355481939) \approx -0.000287970$

$$D''(-0.355481939) \approx 23.51640891$$

$$x^{(3)} = -0.355481939 - \frac{-0.000287970}{23.51640891}$$

$$x^{(3)} = -0.355469693$$

it.4 $D'(-0.355469693) \approx -6.40 \times 10^{-10}$

$$D''(-0.355469693) \approx 23.51630443$$

$$\therefore x^{(4)} = -0.355469693 - \frac{-6.40 \times 10^{-10}}{23.51630443}$$

$$x^{(4)} = -0.35546968$$

Since $|x^{(4)} - x^{(3)}| \approx 2.72 \times 10^{-11} < 10^{-7}$

the method has converged

Newton Point $(6, 0)$

it.1 $D'(0) = -12$

$$D''(0) = 22$$

$$x^{(1)} = 0 - \frac{-12}{22}$$

$$x^{(1)} = 0.54$$

it.2 $D'(0.54) \approx 0.649135988$

$$D''(0.54) \approx 25.57024793$$

$$x^{(2)} = 0.54 - \frac{0.649135988}{25.57024793}$$

$$x^{(2)} = 0.520068167$$

it.3 $D'(0.520068167) \approx 0.004152894$

$$D''(0.520068167) \approx 25.24565078$$

$$x^{(3)} = 0.520068167 - \frac{0.004152894}{25.24565078}$$

$$x^{(3)} = 0.519903668$$

it.4 $D'(0.519903668) \approx 1.68859 \times 10^{-7}$

$$D''(0.519903668) \approx 25.24359788$$

$$x^{(4)} = 0.519903668 - \frac{1.68859 \times 10^{-7}}{25.24359788}$$

$$x^{(4)} = 0.519903661$$

next change is $< 10^{-7}$

$$\therefore x^* = 0.519903661$$

Newton Point $(-8, 0)$

it. $D'(0) = 0 + 0 - 2(-8) = 16$

$$D''(0) = 22$$

$$x^{(1)} = 0 - \frac{16}{22}$$

$$x^{(1)} = -0.7272727$$

it.2 $D'(-0.7272727) \approx -1.538692412$

$$D''(-0.7272727) \approx 28.34710744$$

$$x^{(2)} = -0.7272727 - \frac{-1.538692412}{28.34710744}$$

$$x^{(2)} = -0.672992314$$

it.3 $D'(-0.672992314) \approx -0.025073913$

$$D''(-0.672992314) \approx 27.43502385$$

$$x^{(3)} = -0.67207873$$

it.4 $D'(-0.67207873) \approx -6.74266 \times 10^{-6}$

$$D''(-0.67207873) \approx 27.42027207$$

$$x^{(4)} = -0.672078127$$

it.5 $D'(-0.672078127) \approx -4.87 \times 10^{-15}$

$$x^{(5)} = -0.672078127$$

$$\therefore x^* = -0.672078127$$

PART 13: Golden Section Search

For the parabola $f(x) = x^2 + 5$

& a point $(x_0, 0)$, the squared distance function is:

$$D(x) = (x - x_0)^2 + (x^2 + 5)^2$$

Golden Ratio is: $\phi = \frac{1 + \sqrt{5}}{2} = 1.618034$

fraction used to divide the interval is: $r = 2 - \phi = 0.381966$

For an interval $[a, b]$, the 2 interior points are:

$$x_1 = a + r(b-a)$$

$$x_2 = b - r(b-a)$$

If $D(x_1) < D(x_2) \rightarrow \text{min must lie in } [a, x_2] \therefore \text{new } b = x_2$

$D(x_1) \geq D(x_2) \rightarrow \text{min must lie in } [x_1, b] \therefore \text{new } a = x_1$

Tolerance: $|b-a| < 10^{-7}$

POINT 1: (0,0)

$$D(x) = x^2 + (x^2 + 5)^2$$

initial eval: $x_1 = a + r(b-a)$

$$x_1 = -10 + 0.381966(20)$$

$$x_1 = -2.360680$$

$$x_2 = b - r(b-a)$$

$$x_2 = 10 - 0.381966(20)$$

$$x_2 = 2.360680$$

Squared distance:

$$D(-2.360680) = (-2.36068)^2 + [(-2.36068)^2 + 5]^2$$

$$D(x_1) = 117.357099$$

$$D(2.360680) = (2.36068)^2 + [(2.36068)^2 + 5]^2$$

$$D(x_2) = 117.357099$$

Since $D(x_1) = D(x_2) \Rightarrow a = x_1$

$$\therefore [a, b] = [-2.360680, 10]$$

Iteration 1: $x_1 = -2.360680$

$$x_2 = 10 - 0.381966(10 - (-2.360680))$$

$$x_2 = 5.278640$$

Sqrd dist. $D(-2.360680) = 117.357099$

$$D(5.278640) = 1107.909499$$

$$D(-2.360680) < D(5.278640)$$

$$\therefore \text{min lies in } [-2.360680, 5.278640]$$

Iteration 2: $x_1 = -2.360680$

$$x_2 = -2.360680 + 0.381966(5.278640 - (-2.360680))$$

$$x_2 = 0.557281$$

$$D(0.557281) = 28.512631$$

$$D(-2.360680) = 117.357099$$

$$D(0.557281) < D(-2.360680)$$

$$\therefore [-2.360680, 2.60680]$$

algorithm continues shrinking until $|b-a| < 10^{-7}$

after 40 iterations $x^* \approx 2.70 \times 10^{-8} \approx 0$ & $d \approx 5.0000$

which agrees w/ analytical soln

$$x^* = 0; y^* = 5; d = 5$$

Point 2: $(-4, 0)$

starting interval is: $[a, b] = [-4-10, -4+10] = [-14, 6]$

$$D(x) = (x+4)^2 + (x^2+5)^2$$

$$x_1 = -14 + 0.381966(20)$$

$$x_1 = -6.360680$$

$$x_2 = 6 - 0.381966(20)$$

$$x_2 = -1.639320$$

$$D(-6.360680) = 2072.025$$

$$D(-1.639320) = 64.67$$

$$D(x_2) < D(x_1)$$

$\therefore$ min lies in $[-6.360680, 6]$

it 1. $x_1 = -6.360680$

$$x_2 = 6 - 0.381966(6 - (-6.360680))$$

$$x_2 = 1.278640$$

$$D(-1.639320) = 64.668979$$

$$D(1.278640) = 71.886227$$

$$D(-1.639320) < D(1.278640)$$

$$\therefore [-6.360680, 1.278640]$$

it 2. $x_1 = -1.639320$

$$x_2 = -6.360680 + 0.381966(1.278640 - (-6.360680))$$

$$x_2 = -3.442719$$

$$D(-3.442719) = 284.311076$$

$$D(-1.639320) = 64.668979$$

$$D(-1.639320) < D(-3.442719)$$

$$\therefore [-3.442719, 1.278640]$$

... after 40 iterations

$$x^* \approx -0.355470$$

$$d \approx 6.289845$$

$\therefore$ corresponding point on the parabola is

$$y^* = (-0.355470)^2 + 5$$

$$y^* \approx 5.126359$$

Point 3: $(-8, 0)$

starting interval: $[a, b] = [-8-10, -8+10] = [-18, 2]$

$$D(x) = (x+8)^2 + (x^2+5)^2$$

$$x_1 = -18 + 0.381966(20)$$

$$x_1 = -10.360680$$

$$x_2 = 2 - 0.381966(20)$$

$$x_2 = -5.639320$$

$$D(-10.360680) = 12626.68$$

$$D(-5.639320) = 1359.96$$

$$D(x_2) < D(x_1)$$

$$\therefore [-10.360680, 2]$$

it. 1 $x_1 = -5.639320$

$$x_2 = 2 - 0.381966(2 - (-5.639320))$$

$$x_2 = -2.721360$$

$$D(-5.639320) = 1359.955$$

$$D(-2.721360) = 181.767864$$

$$D(-2.721360) < D(-5.639320)$$

$$\therefore [-5.639320, 2]$$

it. 2 $x_1 = -2.721360$

$$x_2 = -5.639320 + 0.381966(2 - (-5.639320))$$

$$x_2 = -0.917961$$

$$D(-2.721360) = 181.767864$$

$$D(-0.917961) = 84.291861$$

$$D(-0.917961) < D(-2.721360)$$

$$[-5.639320, 2]$$

$$[a, b] = [-2.721360, 2]$$

after 40 iterations

$$x^* \approx -0.672078$$

$$d \approx 9.133420$$

$\therefore$ corresponding point on parabola is:

$$y^* = (-0.672078)^2 + 5$$

$$y^* \approx 5.451689$$

Point 4: (2, 0):

$$[a, b] = [-10, 2+10] = [-8, 12]$$

$$D(x) = (x-2)^2 + (x^2+5)^2$$

$$x_1 = -8 + 0.381966(20)$$

$$x_1 = -0.360680$$

$$x_2 = 12 - 0.381966(10)$$

$$x_2 = 4.360680$$

$$D(-0.360680) = 31.890631$$

$$D(4.360680) = 582.318791$$

$$D(x_1) < D(x_2)$$

$$[-8, 4.36068]$$

iteration 1 $x_1 = -0.360680$

$$x_2 = -8 + 0.381966(4.36068 - (-8))$$

$$x_2 = -3.278640$$

$$D(-0.360680) = 31.890631$$

$$D(-3.278640) = 275.710266$$

$$D(-0.360680) < D(-3.278640)$$

$$\therefore [-3.278640, 4.36068]$$

iteration 2 $x_1 = -0.360680$

$$x_2 = -3.278640 + 0.381966(4.36068 - (-3.278640))$$

$$x_2 = 1.442719$$

$$D(-0.360680) < D(1.442719)$$

$$[-3.278640, 1.442719]$$

$$x^* \approx 0.180745$$

$$d = 5.351306$$

$$y^* = (0.180745)^2 + 5$$

$$y \approx 5.032669$$

Point 5: (6, 0)

$$[a, b] = [6-10, 6+10] = [-4, 16]$$

$$D(x) = (x-6)^2 + (x^2+5)^2$$

$$x_1 = -4 + 0.381966(20)$$

$$x_1 = 3.63932$$

$$x_2 = 16 - 0.381966(20)$$

$$x_2 = 8.36068$$

$$D(3.63932) = 338.44$$

$$D(8.36068) = 5615.727$$

$$D(x_1) < D(x_2)$$

iteration 1 $x_1 = 3.639320$

$$x_2 = -4 + 0.381966(8.36068 - (-4))$$

$$x_2 = 0.721360$$

$$D(3.639320) = 338.44$$

$$D(0.721360) = 58.338$$

$$D(0.721360) < D(3.639320)$$

$$\therefore [-4, 3.63932]$$

iteration 2 $x_1 = 0.721360$

$$x_2 = -4 + 0.381966(3.639320 - (-4))$$

$$x_2 = -1.082039$$

$$D(0.721360) = 58.238416$$

$$D(-1.082039) = 88.234166$$

$$D(0.721360) < D(-1.082039)$$

$$\therefore [-1.082039, 3.639320]$$

$$x^* \approx 0.519904$$

$$d \approx 7.603125$$

$$y^* = (0.519904)^2 + 5$$

$$y^* \approx 5.2703$$

PART 2A: line

assume $\hat{y} = mx + b$

the residual is: $e_i = y_i - \hat{y}_i$

therefore

$$MSE = \frac{1}{4} [(0.5-b)^2 + (1.5-m-b)^2 + (3.5-2m-b)^2 + (3.5-2m-b)^2 + (7.5-3m-b)^2]$$

1st $(0.5-b)^2 = 0.25 - b + b^2$

2nd $(1.5-m-b)^2 = 2.25 + m^2 + b^2 - 3m - 3b + 2mb$

3rd $(3.5-2m-b)^2 = 12.25 + 4m^2 + b^2 - 14m - 7b + 4mb$

4th $(7.5-3m-b)^2 = 56.25 + 9m^2 + b^2 - 45m - 15b + 6mb$

const terms: $0.25 + 2.25 + 12.25 + 56.25 = 71$

m^2 terms: $m^2 + 4m^2 + 9m^2 = 14m^2$

mb : $2mb + 4mb + 6mb = 12mb$

m : $-3m - 14m - 45m = -62m$

b : $-b - 3b - 7b - 15b = -26b$

$$\therefore MSE = \frac{1}{4} (14m^2 + 12mb + 4b^2 - 62m - 26b + 71)$$

$$\frac{\partial MSE}{\partial m} = \frac{1}{4} (28m + 12b - 62)$$

$$28m + 12b - 62 = 0$$

$$14m + 6b = 31$$

$$\frac{\partial MSE}{\partial b} = \frac{1}{4} (12m + 8b - 26)$$

$$12m + 8b - 26 = 0$$

$$6m + 4b = 13$$

$$\begin{cases} 14m + 6b = 31 \\ 6m + 4b = 13 \end{cases}$$

$$\begin{cases} 14m + 6b = 31 \\ 6m + 4b = 13 \end{cases}$$

$$4b = 13 - 6m \xrightarrow{\text{sub}} 14m + 6\left(\frac{13-6m}{4}\right) = 31$$

$$56m + 78 - 36m = 124$$

$$20m = 46$$

$$\therefore m = 2.3$$

$$b = \frac{13 - 6(2.3)}{4} = -\frac{0.8}{4} = -0.2$$

$$\therefore \hat{y} = 2.3x - 0.2$$

Calculating MSE

POINT 1: $x=0, y=0.5$

prediction $\hat{y} = 2.3(0) - 0.2 = -0.2$

error: $e = 0.5 - (-0.2) = 0.7$

$$e^2 = 0.49$$

POINT 2: $x=1, y=1.5$

$\hat{y} = 2.3(1) - 0.2 = 2.1$

$e = 1.5 - 2.1 = -0.6$

$$e^2 = 0.36$$

POINT 3: $x=2, y=3.5$

$\hat{y} = 2.3(2) - 0.2 = 4.4$

$e = 3.5 - 4.4 = -0.9$

$$e^2 = 0.81$$

POINT 4: $x=3, y=7.5$

$\hat{y} = 2.3(3) - 0.2 = 6.7$

$e = 7.5 - 6.7 = 0.8$

$$e^2 = 0.64$$

$$\therefore SSE = 0.49 + 0.36 + 0.81 + 0.64$$

$$SSE = 2.30$$

$$MSE = \frac{2.30}{4}$$

$$MSE = 0.575$$

PART 2B: Parabola

assume $\hat{y} = ax^2 + bx + c$

the MSE is

$$MSE = \frac{1}{4} \sum_{i=1}^4 (y_i - ax_i^2 - bx_i - c)^2$$

$$\therefore MSE = \frac{1}{4} [(0.5 - c)^2 + (1.5 - a - b - c)^2 + (3.5 - 4a - 2b - c)^2 + (7.5 - 9a - 3b - c)^2]$$

$$(0.5 - c)^2 = 0.25 - c + c^2$$

$$(1.5 - a - b - c)^2 = 2.25 + a^2 + b^2 + c^2 - 3a - 3b - 3c + 2ab + 2ac + 2bc$$

$$(3.5 - 4a - 2b - c)^2 = 12.25 + 16a^2 + 4b^2 + c^2 - 28a - 14b - 7c + 16ab + 8ac + 4bc$$

$$(7.5 - 9a - 3b - c)^2 = 56.25 + 81a^2 + 9b^2 + c^2 - 135a - 45b - 15c + 54ab + 18ac + 6bc$$

Adding...

$$MSE = \frac{1}{4} [98a^2 + 36b^2 + 4c^2 + 72ab + 28ac + 12bc - 166a - 62b - 26c + 71]$$

Partial Derivative w.r. to a

$$\frac{\partial MSE}{\partial a} = \frac{1}{4} (196a + 72b + 28c - 166)$$

$$196a + 72b + 28c - 166 = 0$$

$$98a + 36b + 14c = 83 \quad (1)$$

Partial Derivative w.r. to c

$$\frac{\partial MSE}{\partial c} = \frac{1}{4} (8c + 28a + 12b - 26)$$

$$14a + 6b + 4c = 13 \quad (3)$$

$\therefore$ correct normal eqns are:

$$\begin{cases} 98a + 36b + 14c = 83 \\ 36a + 36b + 6c = 31 \\ 14a + 6b + 4c = 13 \end{cases}$$

Partial Derivative w.r. to b

$$\frac{\partial MSE}{\partial b} = \frac{1}{4} (72a + 72b + 12c - 62)$$

$$\frac{\partial MSE}{\partial b} = \frac{1}{4} (72a + 72b + 12c - 62)$$

$$72a + 72b + 12c - 62 = 0$$

$$36a + 36b + 6c = 31 \quad (2)$$

Solve 3 EQNs

$$(1) \quad 98a + 36b + 14c = 83$$

$$(2) \quad 36a + 36b + 6c = 31$$

$$(3) \quad 14a + 6b + 4c = 13$$

$$(1) \times 3 \quad 42a + 18b + 12c = 31$$

$$(2) \times 2 \quad 72a + 28b + 12c = 62$$

$$(42a + 18b + 12c) - (72a + 28b + 12c) = 31 - 62$$

$$-30a - 10b = -23$$

$$\therefore 30a + 10b = 23$$

$$3a + b = 2.3$$

(3) $\times 5$

$$49a + 21b + 14c = 45.5$$

$$(98a + 36b + 14c) - (49a + 21b + 14c) = 83 - 45.5$$

$$49a + 15b = 37.5$$

$$b = 2.3 - 3a$$

$$49a + 15(2.3 - 3a) = 37.5$$

$$4a = 3$$

$$a = 0.75$$

$$b = 2.3 - 3(0.75)$$

$$b = 0.05$$

$$14(0.75) + 6(0.05) + 4c = 13$$

$$4c = 2.2$$

$$c = 0.55$$

$\therefore$ Analytical Best Fit Parabola is:

$$\hat{y} = 0.75x^2 + 0.05x + 0.55$$